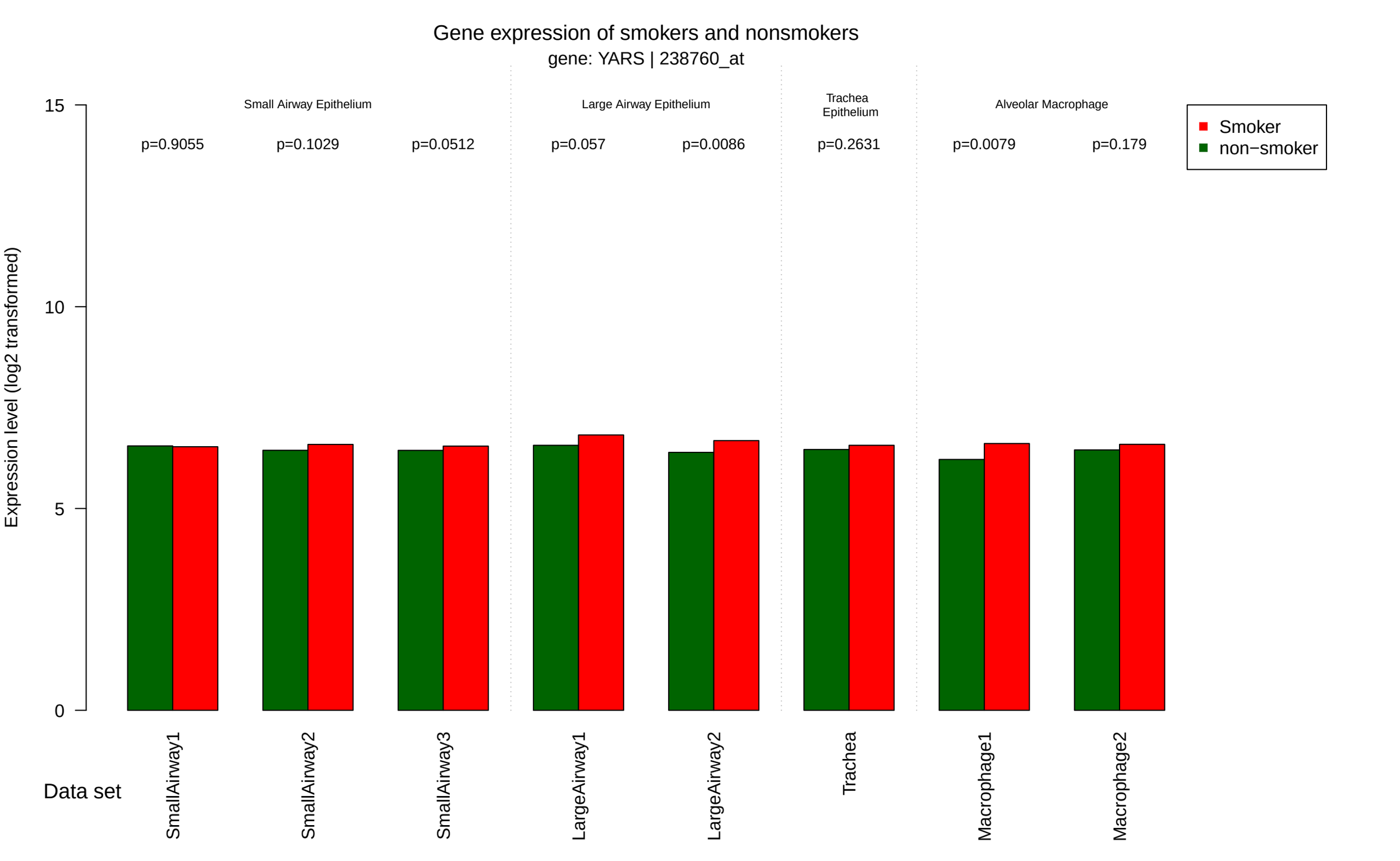
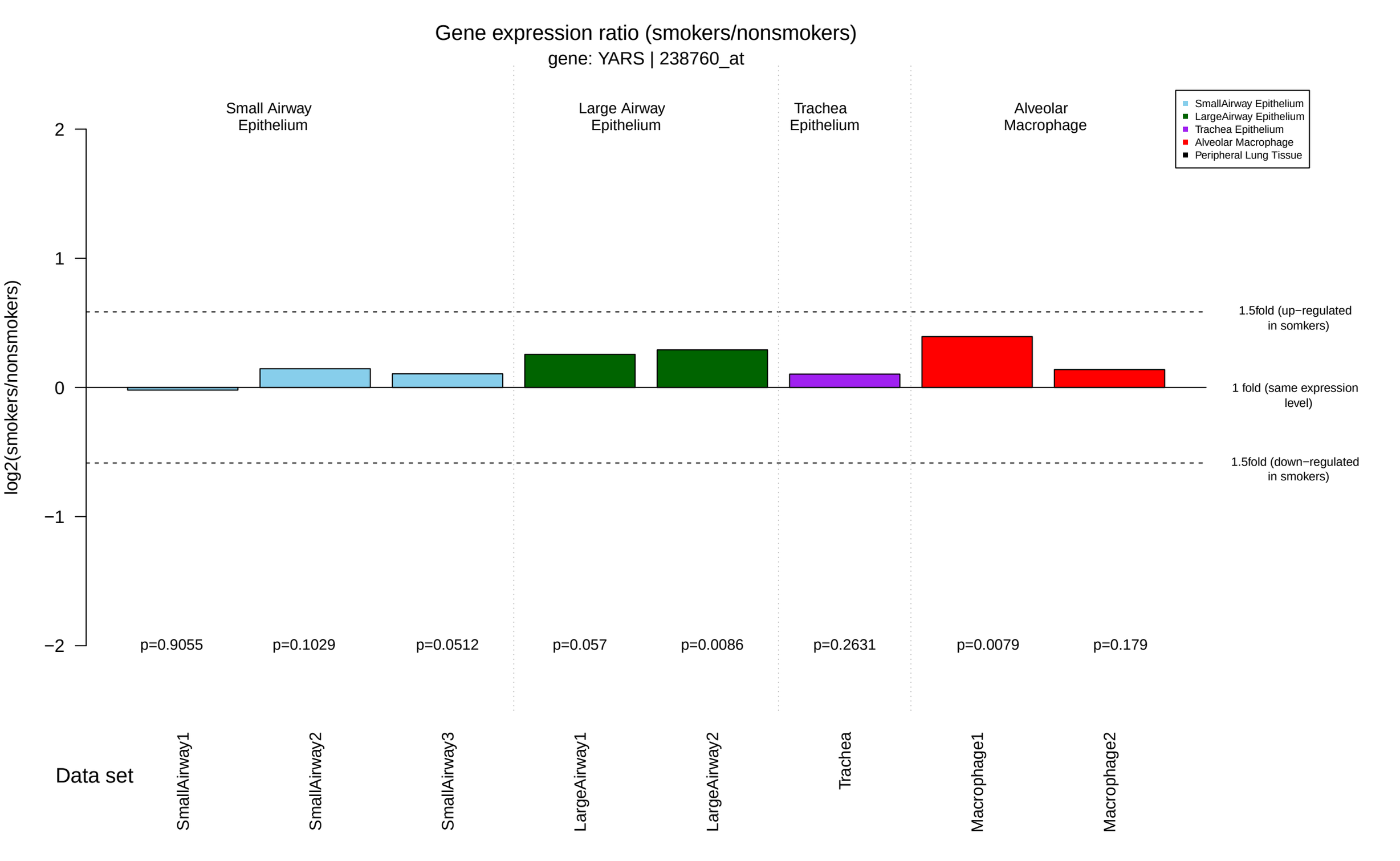
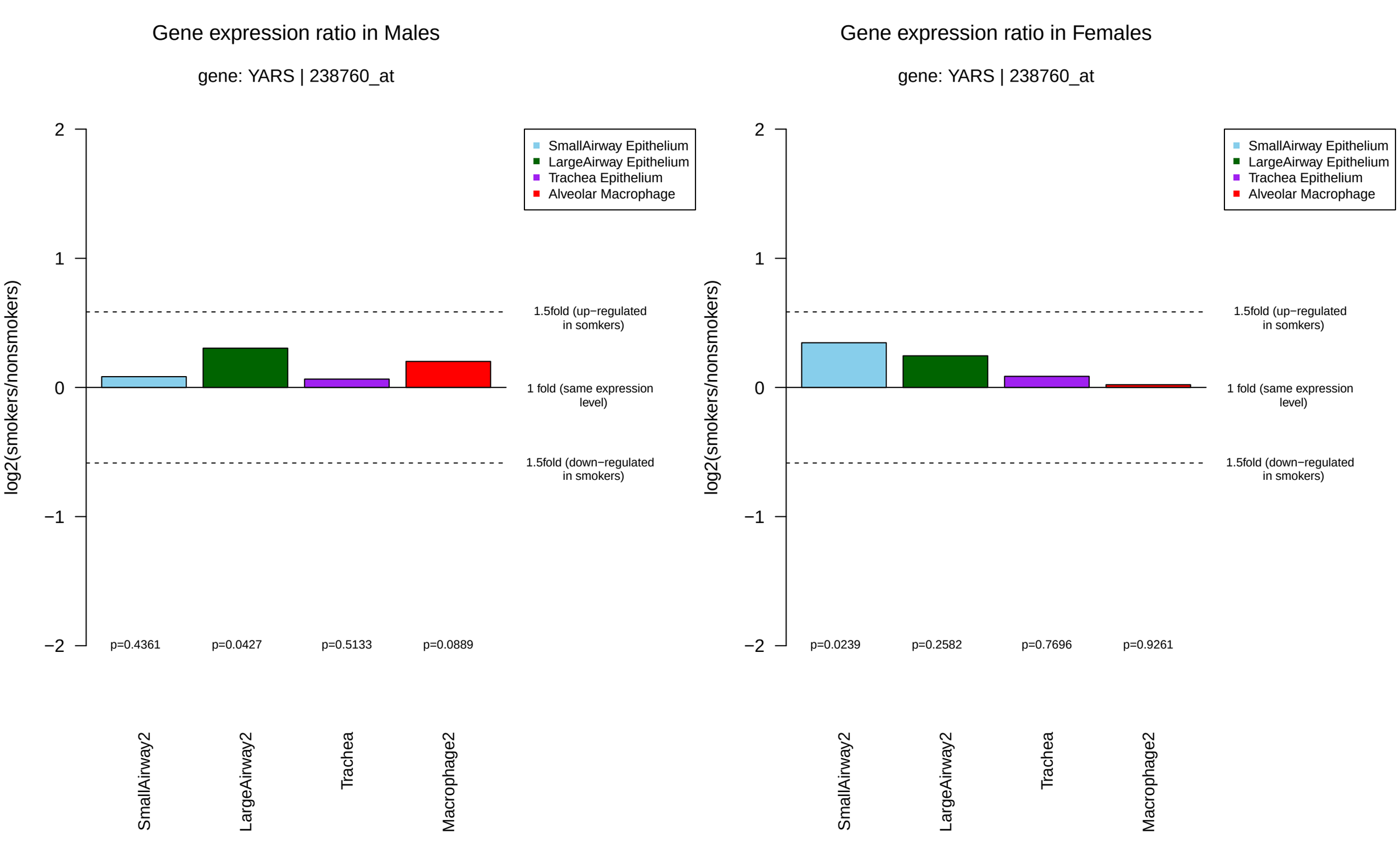
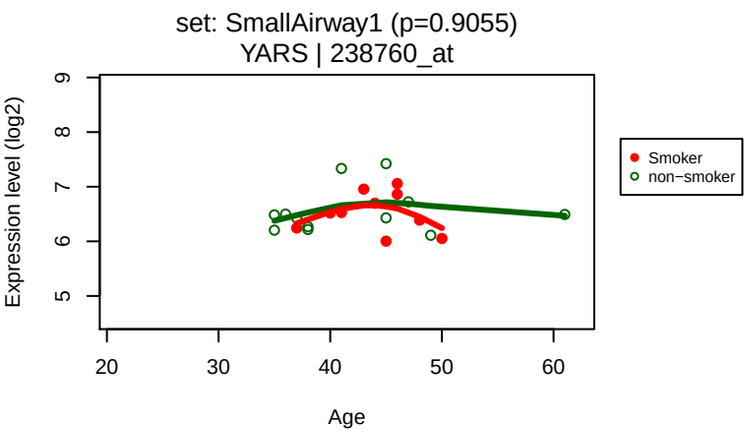
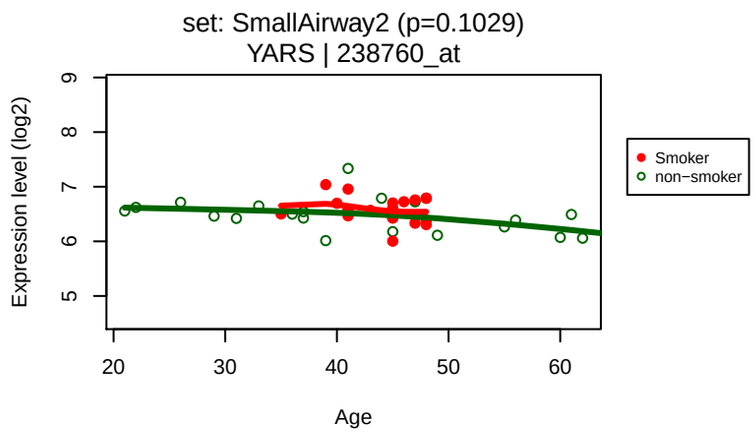
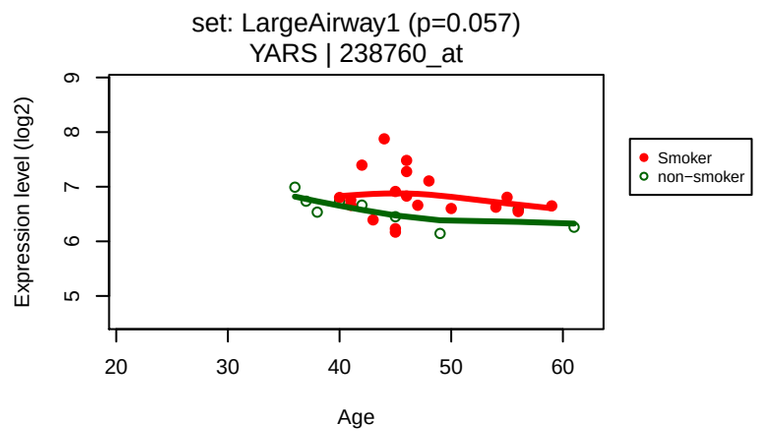

Gene expression of smokers and nonsmokers
gene: YARS | 238760_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9055
p=0.1029
p=0.0512
p=0.057
p=0.0086
p=0.2631
p=0.0079
p=0.179
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

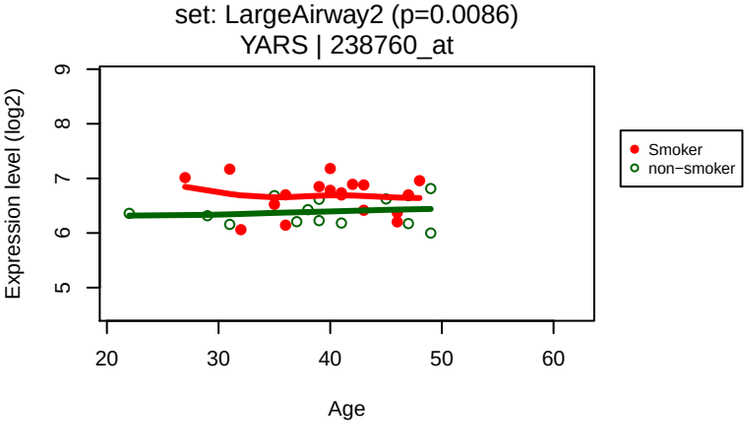

Gene expression ratio (smokers/nonsmokers)
gene: YARS | 238760_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9055
p=0.1029
p=0.0512
p=0.057
p=0.0086
p=0.2631
p=0.0079
p=0.179
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

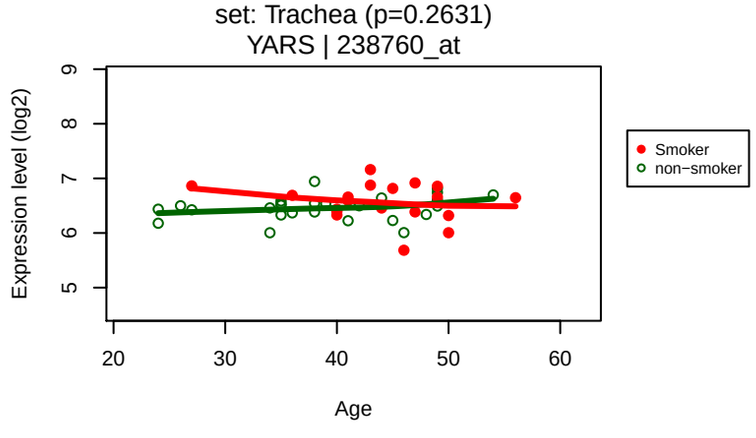

Gene expression ratio in Males
gene: YARS | 238760_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4361
p=0.0427
p=0.5133
p=0.0889
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: YARS | 238760_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0239
p=0.2582
p=0.7696
p=0.9261
SmallAirway2
LargeAirway2
Trachea
Macrophage2

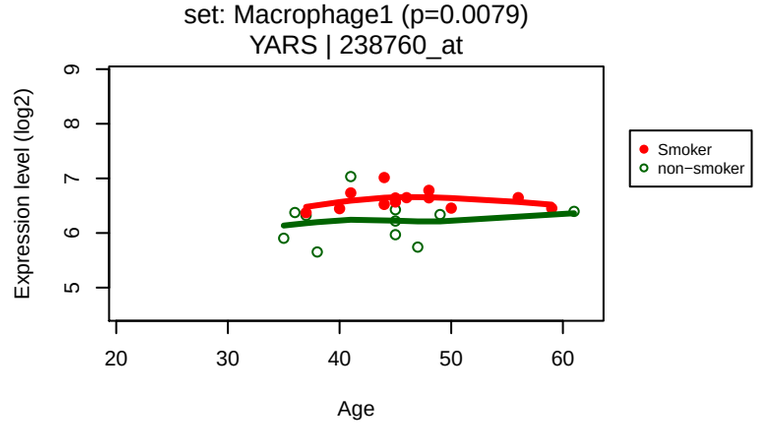

set: SmallAirway1 (p=0.9055)
YARS | 238760_at
Expression level (log2)
Age
Smoker
non−smoker

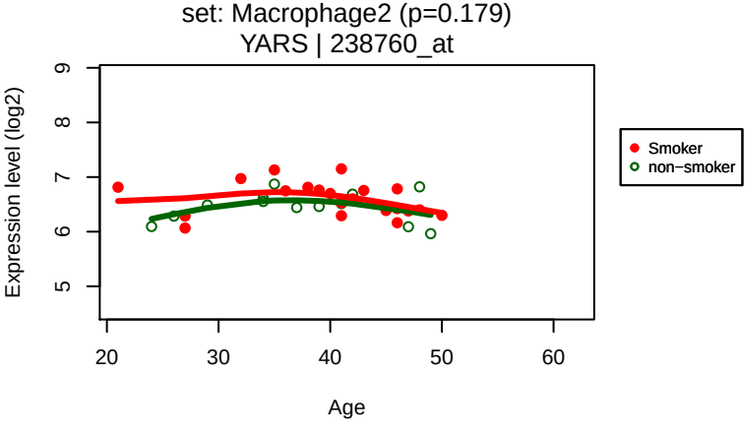

set: SmallAirway2 (p=0.1029)
YARS | 238760_at
Expression level (log2)
Age
Smoker
non−smoker

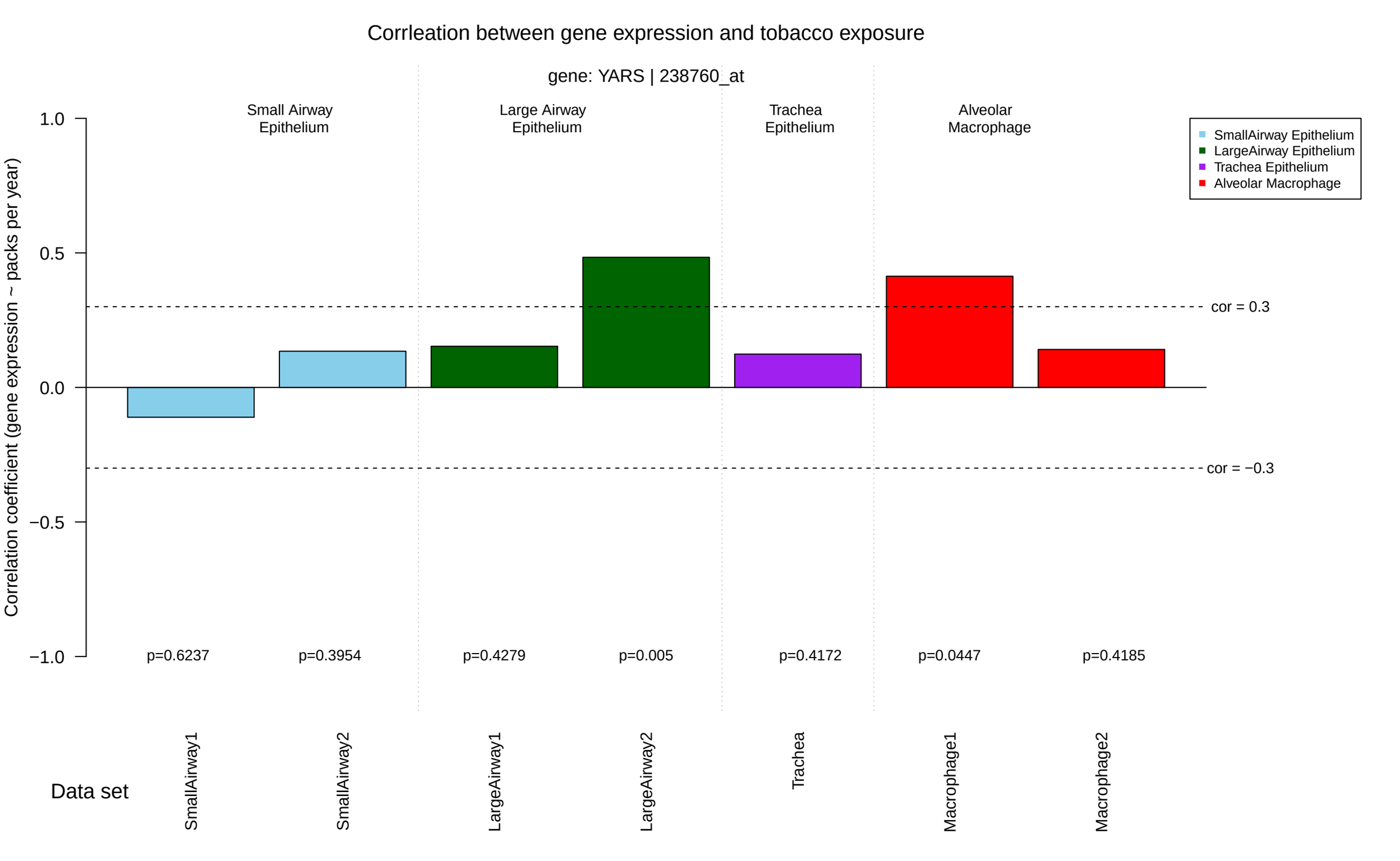

set: LargeAirway1 (p=0.057)
YARS | 238760_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0086)
YARS | 238760_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2631)
YARS | 238760_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0079)
YARS | 238760_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.179)
YARS | 238760_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: YARS | 238760_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6237
p=0.3954
p=0.4279
p=0.005
p=0.4172
p=0.0447
p=0.4185
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2